SINGAPORE-CAMBRIDGE GENERAL CERTIFICATE OF EDUCATION ORDINARY LEVEL

ADDITIONAL MATHEMATICS

Paper 2

Time: **2 hours 15 minutes** Marks: **90**

READ THESE INSTRUCTIONS FIRST

- Write your name, index number, and class clearly in the spaces provided.
- Write in dark blue or black pen. You may use an HB pencil for any diagrams or graphs.
- Do not use staples, paper clips, glue or correction fluid.
- Answer **all** questions.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
- The use of an approved scientific calculator is expected, where appropriate.
- You are reminded of the need for clear presentation in your answers.

MATHEMATICAL FORMULAE

1. ALGEBRA

Quadratic Equation

For the equation $ax^2 + bx + c = 0$, $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Binomial expansion

$(a+b)^n = a^n + \binom{n}{1}a^{n-1}b + \binom{n}{2}a^{n-2}b^2 + \cdots + \binom{n}{r}a^{n-r}b^r + \cdots + b^n$, where n is a positive integer and $\binom{n}{r} = \dfrac{n!}{r!(n-r)!} = \dfrac{n(n-1)\ldots(n-r+1)}{r!}$

2. TRIGONOMETRY

Identities

$$\begin{aligned}
\sin^2 A + \cos^2 A &= 1 \\
\sec^2 A &= 1 + \tan^2 A \\
\operatorname{cosec}^2 A &= 1 + \cot^2 A \\
\sin(A \pm B) &= \sin A \cos B \pm \cos A \sin B \\
\cos(A \pm B) &= \cos A \cos B \mp \sin A \sin B \\
\tan(A \pm B) &= \frac{\tan A \pm \tan B}{1 \mp \tan A \tan B} \\
\sin 2A &= 2 \sin A \cos A \\
\cos 2A &= \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A \\
\tan 2A &= \frac{2\tan A}{1 - \tan^2 A}
\end{aligned}$$

Formulae for $\triangle ABC$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc\cos A$$

$$\Delta = \frac{1}{2}ab\sin C$$

Question 1 **[6 marks]**

(a) Solve the equation $2\sqrt{2x+3} - x = 3$. [3]

(b) Express $\dfrac{7\sqrt{5}-1}{3\sqrt{5}+4}$ in the form $p + q\sqrt{5}$, where p and q are rational numbers. [3]

[Total: 6]

Question 2 **[8 marks]**

(a) Find the quotient and remainder when $6x^3 + 7x^2 - x + 5$ is divided by $2x - 1$. [2]

(b) Hence, express $\dfrac{6x^3 + 7x^2 - 9x + 2}{(2x-1)(x^2+1)}$ in partial fractions. [6]

[Total: 8]

Question 3 **[8 marks]**

(a) Find the term independent of x in the binomial expansion of $\left(2x - \frac{1}{4x}\right)^{10}$. [3]

(b) In the expansion of $(1 + kx)(2 - x)^6$ in ascending powers of x, where k is a non-zero constant, the coefficient of x^2 is 28.

(i) Find the value of k. [3]

(ii) Using this value of k, determine the coefficient of x^3. [2]

[Total: 8]

Question 4 **[8 marks]**

(a) Solve the equation $\log_3(x-2) + \log_3(x+4) = 1 + \log_3 5$. [4]

(b) Solve the simultaneous equations:

$$2^{2x+1} \times 4^y = 32$$
$$\log_2(x+3y) = 2$$

[4]

[Total: 8]

Question 5 **[9 marks]**

The variables x and y are related by the equation $y = ab^x$, where a and b are positive constants.

When $\lg y$ is plotted against x, a straight line is obtained which passes through the points $(2, 1.20)$ and $(5, 2.55)$.

(a) Express $y = ab^x$ in the linear form $Y = mX + c$, clearly stating expressions for Y, X, m and c. [2]

(b) Calculate the value of $\lg a$ and the value of $\lg b$. [3]

(c) Hence, determine the value of a and of b, giving each answer correct to 3 significant figures. [2]

(d) Find the value of x when $y = 50$. [2]

[Total: 9]

Question 6 **[10 marks]**

(a) State the amplitude and period of the function $f(x) = 3 - 2\cos(4x)$. [2]

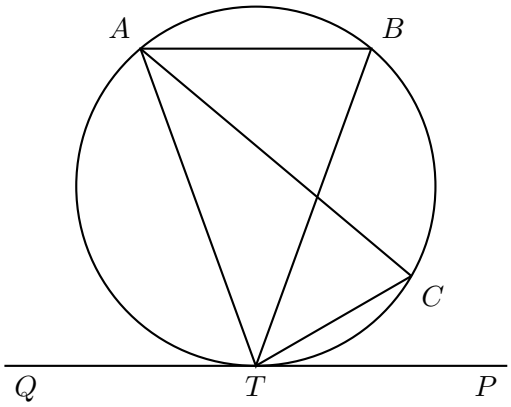
(b) Express $4\cos\theta - 3\sin\theta$ in the form $R\cos(\theta + \alpha)$, where $R > 0$ and $0° < \alpha < 90°$. [3]

(c) Hence, solve the equation $4\cos\theta - 3\sin\theta = 2.5$ for $0° \leq \theta \leq 360°$. [3]

(d) State the minimum value of $\dfrac{1}{8 + 4\cos\theta - 3\sin\theta}$ and find the value of θ in the range $0° \leq \theta \leq 360°$ at which this minimum occurs. [2]

[Total: 10]

Question 7 **[8 marks]**



In the diagram, PTQ is a tangent to the circle at the point T. A, B and C are points on the circle such that the chord AB is parallel to the tangent PTQ. The line TC intersects the chord AB at the point M.

(a) Prove that $\triangle TAB$ is an isosceles triangle with $TA = TB$. [4]

(b) Given that TC bisects $\angle ATB$, prove that $\triangle TMA$ is similar to $\triangle TAC$. [4]

[Total: 8]

Question 8 **[11 marks]**

A curve has the equation $y = \dfrac{\ln(2x-1)}{2x-1}$ for $x > \dfrac{1}{2}$.

(a) Show that $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{2\big(1-\ln(2x-1)\big)}{(2x-1)^2}$. [3]

(b) Find the exact coordinates of the stationary point of the curve and determine the nature of this stationary point. [4]

(c) Find the equation of the normal to the curve at the point where $x = 1$. [4]

[Total: 11]

Question 9 **[10 marks]**

A particle P moves in a straight line such that, t seconds after passing a fixed point O, its velocity v m/s is given by

$$v = 6 - 2\mathrm{e}^{0.5t}.$$

(a) Find the initial velocity and initial acceleration of P. [2]

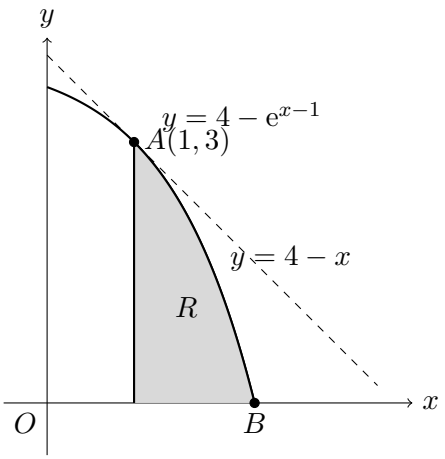
(b) Find the value of t when P is instantaneously at rest, giving your answer in exact logarithmic form. [2]

(c) Find the displacement of P from O when it is instantaneously at rest. [3]

(d) Calculate the total distance travelled by P in the first 3 seconds of its motion. [3]

[Total: 10]

Question 10 **[12 marks]**



The diagram shows part of the curve $y = 4 - e^{x-1}$. The curve intersects the x-axis at the point B and passes through the point $A(1, 3)$. The tangent to the curve at A meets the y-axis at the point $(0, 4)$.

(a) Find the exact x-coordinate of the point B. [2]

(b) Show that the equation of the tangent to the curve at the point A is $y = 4 - x$. [3]

(c) Find the exact area of the shaded region R, bounded by the curve, the line $x = 1$ and the x-axis. [4]

(d) Hence, determine the exact area of the region bounded by the curve, the tangent at A, and the y-axis. [3]

[Total: 12]